

Corrosion + Rust Removal

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Find the battery nobody wanted, then run it backwards

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 5 — Corrosion and Electrolytic Rust Removal

Hook: Rust is a battery you did not want. Today we measure the voltage that drives it, make iron rust faster on purpose, protect it, and open the cells that spent the night running the reaction backwards.

Learning outcomes

- **Predict** the voltage of a metal pair from the electrochemical series, then **measure** it and account for the gap
- Measure the **short-circuit current** of a galvanic couple and convert it, via Faraday's law, into a **corrosion rate in grams per year**
- Explain galvanic corrosion and sacrificial protection, and identify which metal in a pair is at risk
- Interpret the overnight **electrolytic rust removal** cells started in Session 4
- Place all five sessions on a single axis: **spontaneous chemistry making electricity**, or **electricity forcing chemistry**

Session plan (90 min)

Time	Activity	Reference
0–15 min	Part A — Reveal the derusted nails and the corrosion jars	experiment.md — Part A
15–30 min	Activity series, galvanic couples, sacrificial protection	lecture.md §3
30–55 min	Part B — Metal-pair cells: predict, measure, then measure the current	experiment.md — Part B
55–68 min	Part C — Corrosion jars A / B / C	experiment.md — Part C
68–78 min	Part D — The week on one table	experiment.md — Part D
78–90 min	Capstone challenge — applied problems, then the closing	lecture.md §5

Key reactions

Rusting, as two half-reactions happening on the same piece of iron:

Anodic sites: $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$

Cathodic sites: $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$

Fe^{2+} is then further oxidized and precipitates as hydrated iron oxides — rust.

Galvanic couple (Fe touching Cu in saltwater):

Iron becomes the anode of a real cell and corrodes faster.

Sacrificial protection (Fe touching Zn):

$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ instead. Iron is now the cathode and is protected.

Electrolytic derusting:

A power supply forces the iron to be the cathode. Oxide is reduced and lifted; H_2 evolves.

Multi-day setup this session depends on

Demo	Must be started	Revealed
Corrosion jars A / B / C	Session 1 — see shared/multi-day-setup.md	Session 5
Electrolytic rust removal	Session 4 , end of class	Session 5, opening

If the jars were not started on Day 1, read the fallback in preparation.md before class rather than during it.

Status

- ☐ Derusting cells confirmed still powered this morning
- ☐ "Before" photos located and ready to display
- ☐ Corrosion jars A / B / C available (or fallback plan chosen)
- ☐ Metal strips cleaned: Zn, Fe, Cu, Al, and Mg if available
- ☐ 5–10% saltwater mixed and labeled "**corrosion demos only**"
- ☐ Capstone worksheets printed

Session 5 — Lecture Notes: Corrosion, Galvanic Couples, and Reversing Rust

Audience: instructors / TAs — use this as your teaching script. **Students** see the talking points, do the Part B predictions, and finish with the \$5 capstone.

Deep-dive chemistry, potentials, coaching traps: instructor.md. Bench procedure: experiment.md.

1. Where the week has got to

This is the synthesis day. Do not treat §1 as revision for its own sake — every idea listed here is load-bearing in today's lab.

Session	The idea we need today
1 — Fruit battery	Two different metals in an electrolyte make a spontaneous cell with a measurable voltage
2 — Copper plating	Electricity can force a reaction that would not happen on its own; polarity decides everything
3 — Silver plating	$Q = It$, $n = Q/nF$. Current is a rate of chemistry, and we can count atoms with it
4 — Electrolysis	A driven cell splits water; hydrogen appears at whatever we make negative

The framing sentence for the whole day:

"Day 1 we made a battery on purpose. Today we are going to find one we did not want — and it has been eating iron for as long as there has been iron."

What is new today

- The same galvanic cell from Day 1, but now it is a **problem** instead of a demonstration
- We **predict** a cell voltage from a table before measuring it — and then account for why the measurement falls short
- Current is reinterpreted one last time: in Session 3 it was a plating rate, in Session 4 a gas rate, and today it is a **corrosion rate in grams per year**
- Everything the week has covered collapses onto one axis: spontaneous, or driven

2. Class plan (90 min)

Time	Block	What you do	Student product
0–15 min	Part A — the reveal	Open the cells; nail, control, before-photo, side by side	Observation sheet
15–30 min	Teach today's concepts	§3; activity series on the board	Notes / vocab
30–55 min	Part B — predict, measure, measure again	Predictions before meters are handed out	Prediction vs measurement table + corrosion rate

Time	Block	What you do	Student product
55–68 min	Part C — corrosion jars	Rank A/B/C; link to their own Part B voltages	Jar sheet
68–78 min	Part D — week table	Groups fill from memory, then compare	Completed week table
78–90 min	Capstone + closing	\$5	Capstone worksheet

Pacing tip: the reveal is the best fifteen minutes of the week and it is tempting to let it run. Don't — Part B needs its full twenty-five minutes, because the predict-then-measure gap is where the real chemistry of the day lives.

3. Detailed explanations to give students

3.1 Open with the reveal, not with talking

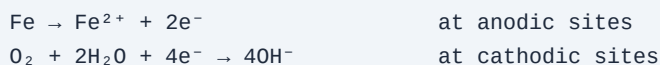
Hold up the treated nail and the control together, and say nothing for a moment. Then:

"Same nail, same solution, same jar, same night. The only difference is that one of them had wires on it. What did the electricity do?"

Take answers before correcting any of them. Then park the question — you will come back to it in §3.5 — and go and build the corrosion story first, because derusting only makes sense once they know what rusting is.

3.2 Rust is not dirt

The idea to dislodge, and most students hold it: that rust is a substance that lands on iron, like dust.

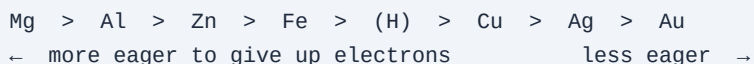


Then Fe^{2+} is oxidized further and precipitates as hydrated iron oxides.

Rust is the iron. It is the same atoms, further along, having given up electrons. Nothing arrived — something left. This is why a rusted-through bolt has lost material, and why rust weighs *more* than the iron it came from, not less.

Note what the cathodic half-reaction requires: **oxygen and water, both**. Iron does not rust in dry air, and does not rust in oxygen-free water. Ask why a car rusts at the wheel arches and a bicycle rusts where the paint chipped.

3.3 The activity series, and what it is really telling you



Metal	E° (V)	Where you meet it
Mg	–2.37	Sacrificial anodes in water heaters
Al	–1.66	Everything — because of its oxide film, see §3.6

Metal	E° (V)	Where you meet it
Zn	-0.76	Galvanizing; ship hull anodes
Fe	-0.44	The metal we are trying to save
Cu	+0.34	Plumbing, wiring — and a menace next to steel
Ag	+0.80	Session 3

The rule, and it is the only one they need to memorise: put two of these in contact in an electrolyte, and the one **higher in the list corrodes**. The other is protected.

Then the payoff:

$$\text{Predicted voltage} = E^\circ(\text{protected metal}) - E^\circ(\text{corroding metal})$$

This is Session 1's fruit battery with the mystery removed. The lemon was never the point; the two different metals were.

3.4 The two consequences, which are the same fact

Galvanic corrosion — the bad case. Iron touching copper in saltwater. Copper is nobler, so iron becomes the anode of a real cell and corrodes *faster than it would alone*. Jar B. This ruins real structures: a copper pipe threaded into a galvanized fitting, a steel bolt in an aluminium panel, a bronze propeller on a steel shaft.

Sacrificial protection — the good case. Iron touching zinc. Now zinc is the more active metal, so zinc becomes the anode and iron becomes the cathode. Jar C. The iron is protected not by being covered but by being **electrically fed** — which is why a scratch in galvanized steel does not start rusting, while a scratch in painted steel does.

Ask this and let them sit with it:

"Session 3, we silver-plated a copper coin. Suppose that silver layer gets scratched. Which metal corrodes now?"

The copper. It is more active than silver, so the base metal underneath becomes the anode — and worse, it is a **small anode wired to a large cathode**, so the attack concentrates into a pit. A noble coating protects only while it is perfect. A sacrificial coating protects even when it is not. That is the whole difference between chrome plating and galvanizing, and it lands hard on students who assumed plating is plating.

3.5 Now come back to the nail

Rusting made iron the anode. So to reverse it, **make the iron the cathode** — and since that is not what it wants to do, you have to pay for it with a power supply. Session 4's driven cell, pointed at rust.

	Rusting in jar B	Derusting overnight
Iron is the	Anode — oxidized	Cathode — reduced
Driven by	Difference in metal activity	External power supply
Direction	Spontaneous, free, unstoppable	Forced, costs energy
Electrolyte	Saltwater	Washing soda, no chloride
Partner metal	Copper survives	The anode is consumed

And the polarity is the entire experiment. Reverse the clips and you dissolve the nail rather than clean it. Same wires, same solution, same battery — opposite outcome. Nothing else this week makes the point so cheaply.

Then be honest about the mechanism, because students will ask: the current does not turn rust back into shiny steel atom for atom. Some oxide is reduced; hydrogen bubbling at the surface mechanically lifts the rest; loose material falls into the bath as the sludge they can see. Metal already lost to rust is not coming back. Derusting saves what remains — it does not undo history.

3.6 Why the measurements will disagree with the table

Warn them **before** Part B, so the gap is a finding and not a disappointment.

1. **The table assumes 1 M solutions of each metal ion.** Our jar has essentially no Zn^{2+} or Cu^{2+} in it at all.
2. **The copper is not doing what the table says.** In an open jar of saltwater, the reaction at the copper is oxygen reduction — $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$ — not $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$. We are reading the wrong line of the table, deliberately, because it still gets the *direction* right.
3. **Aluminium lies.** Predicted at -1.66 V it should be ferocious. It will probably read below iron, because it is wearing an oxide film only a few nanometres thick that is chemically tough and self-repairing.

That third one is worth dwelling on: aluminium is thermodynamically one of the most reactive metals in daily use, and we make drink cans and aeroplanes out of it. The table tells you what *wants* to happen. It never tells you how fast — and engineering lives in that gap.

3.7 Corrosion as a rate

Last use of Faraday's law this week, and the most sobering:

$$\text{mass of zinc lost per second} = I \times 65.4 / (2 \times 96,485)$$

A steady 1 mA consumes about 11 grams of zinc per year.

Measure the short-circuit current of the Zn-Cu jar and you have measured a corrosion rate. Scale it up and you have sized a ship's sacrificial anode. This is not an analogy for real engineering — *it is* the calculation real engineers do.

3.8 Safety (brief)

- Salt is allowed today, and today only, and only in the corrosion jars
- Disconnect before touching any powered cell
- Rusty hardware is sharp; forceps or gloves
- Wash hands after handling saltwater and metals

4. Quiz questions

Cold call or a 3-minute written check, before Part B. Answers in *italics*.

Q1. Rust weighs more than the iron it formed from. Why? *The iron is still there, combined with oxygen and water. Nothing left; oxygen joined.*

Q2. Iron needs two things to rust. Name both. *Water and oxygen. Remove either and rusting stops — which is why oiled tools and submerged-in-boiled-water samples survive.*

Q3. Zinc and iron are in contact in saltwater. Which corrodes, and how do you know? *Zinc — it is more active, $E^\circ = -0.76$ V versus -0.44 V for iron.*

Q4. Predict the voltage of a Zn–Cu cell from the table. $0.34 - (-0.76) = +1.10$ V.

Q5. Your measured Zn–Cu voltage is 0.95 V, not 1.10 V. Has something gone wrong? *No. The table's conditions (1 M ions, 25 °C) are not our conditions, and the copper electrode is actually reducing oxygen rather than copper ions.*

Q6. In the derusting cell, which terminal did the rusty nail go to, and why that one? *Negative. It makes the iron the cathode, so it is reduced instead of oxidized.*

Q7. What would have happened had we connected the nail to positive? *It would have become the anode and dissolved — accelerated corrosion, in a jar.*

Q8. Why washing soda and not salt in the derusting bath? *Chloride would be oxidized at the anode to chlorine gas, and it drives pitting.*

Q9. Jar A has only one metal in it. Why does that nail rust at all? *A single piece of iron already has anodic and cathodic regions — impurities, stress, scratches, and differences in oxygen access. Two metals only make it worse and more predictable.*

Q10. A galvanized bucket gets scratched down to the steel. Does it start rusting there? *Not straight away. The surrounding zinc is still electrically connected and still more active, so it keeps protecting the exposed steel.*

5. Capstone challenge (78–90 min)

Print this. Groups of 2–3, then discuss as a class. Section C is the part worth the time.

Section A — The week in one line each

1. Which session **made** electricity from chemistry?
2. Which sessions **spent** electricity to force chemistry?
3. Which session **moved metal atoms** from one place to another?
4. Which session **split a molecule**?
5. In every single experiment: where did the **electrons** travel, and where did the **ions** travel?

Section B — Calculations

B1. Your Zn–Cu jar drew a short-circuit current of 4 mA. Using $1 \text{ mA} \approx 11 \text{ g}$ of zinc per year, how much zinc would that couple consume in a year?

B2. A derusting cell ran overnight at 0.4 A for 12 hours. (a) How much charge passed? (b) How many moles of electrons is that? (c) If all of that made hydrogen, what volume of H_2 was produced? (24.45 L/mol) (d) Why does that answer make "keep the jar open and the room ventilated" a rule rather than a suggestion?

B3. A water electrolysis cell runs at 0.25 A for 8 minutes. How many mL of H_2 and of O_2 ? (Use $\approx 7.6 \text{ mL/min}$ per amp of H_2 .)

Section C — Applied scenarios

C1. A boatyard fits **steel** screws into an **aluminium** hull. Six months later something is badly corroded. Which metal, and why? What would you do instead?

C2. A plumber joins a **copper** pipe directly to a **galvanized steel** pipe. Which side fails, and where exactly?

- C3.** A domestic water heater contains a **magnesium rod** that must be replaced every few years. What is it for, and what starts happening once it is fully consumed?
- C4.** A friend plans to derust an antique tool using **saltwater** because it is cheaper. Give them two specific reasons not to.
- C5.** A silver-plated copper spoon gets scratched through the silver. Which metal corrodes — and why is a scratch in silver plating more dangerous than a scratch in galvanizing?
- C6.** You are asked to protect a wrought-iron gate that stands outdoors for thirty years. Propose a strategy, name the electrochemistry behind each element of it, and say what maintenance it needs.

Answer key

Section A. (1) Session 1, and jar B in Session 5 — corrosion is a battery discharging for free. (2) Sessions 2, 3, 4, and the derusting cell. (3) Sessions 2 and 3, and arguably corrosion, which moves iron into solution. (4) Session 4. (5) Electrons through the metal and wires; ions through the solution. Always both, always in a loop.

B1. $4 \text{ mA} \times 11 \text{ g/year per mA} \approx \mathbf{44 \text{ g of zinc per year}}$.

B2. (a) $Q = 0.4 \times 43,200 = \mathbf{17,280 \text{ C}}$. (b) $17,280 / 96,485 = \mathbf{0.179 \text{ mol e}^-}$. (c) $0.0895 \text{ mol H}_2 \times 24.45 \text{ L/mol} \approx \mathbf{2.2 \text{ litres}}$. (d) Two litres of hydrogen accumulated overnight in a sealed container is an explosive hazard; in an open jar in a ventilated room it disperses harmlessly. The rule follows from the arithmetic.

B3. $\text{H}_2: 7.6 \times 0.25 \times 8 = \mathbf{15.2 \text{ mL}}$. O_2 : half that, **7.6 mL**.

C1. The **aluminium hull** corrodes, and it concentrates around each screw. Aluminium is far more active than steel, so it becomes the anode — and a large anode area would be survivable, but here you have a large aluminium anode feeding small steel cathodes, so the damage spreads around every fastener. Fix: aluminium or compatible fasteners, insulating washers and sleeves, or a sealant that breaks the electrical path. Accept any answer that identifies aluminium as anodic and proposes breaking either the electron path or the electrolyte path.

C2. The **galvanized steel** side, right at the joint. Zinc goes first, then the steel underneath. Real plumbing codes require a dielectric union between copper and galvanized steel precisely for this. Credit for noting that the damage is worst *at* the junction, where the two metals and the water all meet.

C3. It is a **sacrificial anode**: magnesium is far more active than the steel tank, so it corrodes preferentially and keeps the tank cathodic. Once it is consumed the tank itself becomes the anode and starts to corrode from the inside — which is why the rod is cheap maintenance and the tank is an expensive failure.

C4. First, chloride is oxidized at the anode to **chlorine gas**, in a bucket, indoors. Second, chloride drives **pitting** — it breaks down passive films and attacks locally, so even where it does clean, it damages the surface it is meant to save. Washing soda costs almost nothing and does neither.

C5. The **copper** corrodes, because it is more active than silver. The danger is geometry: a scratch exposes a **tiny anode** connected to a **large cathode** of intact silver, so all the corrosion current concentrates into that one small spot and drills a pit. Galvanizing is the opposite arrangement — the coating itself is the anode, so a scratch is protected by the surrounding zinc instead of being attacked by it. A noble coating must be perfect to work; a sacrificial coating does not.

C6. Look for a **layered** answer rather than one silver bullet: a barrier (paint or powder coat) to keep water and oxygen off, ideally over a **galvanized** base so that any breach in the paint still has sacrificial protection underneath; attention to **crevices and joints**, where water sits and oxygen is scarce and corrosion concentrates; drainage so the base does not stand in a puddle; and inspection every few years, with touch-up before rust gets under the coating and lifts it. Strong answers mention that paint is *not* electrochemical — it works by denying the reaction its reactants — while galvanizing is, and that combining the two mechanisms is exactly why it is done that way in practice.

6. Closing

Put the week's six experiments on the board in one row, and draw a single arrow under all of them.

"Every experiment this week was the same experiment. Electrons leaving something and arriving somewhere else. On Monday we let it happen and measured the voltage. On Tuesday and Wednesday we paid for it and got silver and copper where we wanted them. On Thursday we paid for it and pulled water apart. Today we found the same reaction running without permission, eating iron — and last night we paid to run it backwards.

The chemistry never changed. All that ever changed was who was paying."

7. Vocabulary — full week review

- [] Oxidation / reduction
 - [] Anode / cathode, and how the signs differ between galvanic and electrolytic cells
 - [] Electrolyte; supporting electrolyte; spectator ion
 - [] Galvanic cell vs electrolytic cell; spontaneous vs driven
 - [] Activity series; standard reduction potential
 - [] Faraday's constant; current efficiency
 - [] Sacrificial anode; cathodic protection; passivation
 - [] Energy carrier vs energy source
-

8. Materials you need in hand while teaching

- The treated nail, the control nail, and the printed before-photo
- Jars A, B and C, arranged where everyone can see them
- Activity series written large enough to stay on the board all session
- A galvanized nail and a plain iron nail, to pass around
- A photo of a ship's hull anode or a water-heater rod
- Printed E° prediction tables with the predictions column **blank**
- Printed capstone worksheets

Session 5 — Instructor Notes: Corrosion and Rust Removal Deep Dive

Audience: instructors and TAs only. Pair with lecture.md and experiment.md.

Why this session exists

Four days of controlled electrochemistry, all of it something we did on purpose. Today the same chemistry turns up **uninvited**, doing billions of dollars of damage a year, and the students discover they already have every tool needed to explain it and to fight it.

Your job is to make **activity series**, **galvanic couples**, and **driven versus spontaneous** land as one idea rather than three demos. The organising question for the whole session: *is this reaction happening because it wants to, or because somebody is paying for it?*

The reveal — protect it, then get out of it

The overnight cells are the strongest visual of the week. Two things make or break it:

The control. A rusty nail sitting in alkaline solution overnight changes appearance all by itself. Without the unpowered control in the same solution, you have shown the students a nail that looks different from a photo, and a sharp student will say so. With the control, you have an experiment.

Disconnecting first. Every time, before any hand goes near the jar. Make it visible and make it ritual — a TA disconnects, then the nail comes out.

Expect this to want to run long. Cap it at fifteen minutes. Part B is where the chemistry of the day actually lives.

What "cleaning" means, honestly

Be straightforward, because the students will push:

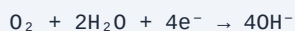
- Some iron oxide at the surface is genuinely electrochemically reduced.
 - A great deal of the visible effect is **mechanical**: hydrogen evolving at the surface undermines and lifts loosely adherent oxide, which falls into the bath as the sludge everyone can see.
 - Metal already lost to rust is gone. Derusting recovers the surface, not the missing steel.
 - The nail comes out grey-black and needs brushing and drying. It is also now bare, active steel with no oxide layer at all, so it will flash-rust within hours if left wet. That is a genuinely satisfying detail to end on — you cleaned it so thoroughly that it is now more vulnerable than when you started.
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Corrosion chemistry you should own

Rust, properly



anodic dissolution



cathodic oxygen reduction, neutral water

Fe^{2+} oxidizes further and precipitates as hydrated oxides and oxyhydroxides — the FeOOH / $\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$ family. The exact mineralogy is genuinely complicated and depends on chloride, pH and oxygen; do not go there with this age group. What they need is: **rust is oxidized iron, not a deposit that landed on iron.**

Note the cathodic step needs **both** water and oxygen. This explains: oiled tools survive, fully submerged deoxygenated iron survives, and the waterline of a jar is the worst place of all because it has generous access to both.

Why a lone nail rusts (jar A)

Local cells on a single piece of metal, arising from impurities and grain boundaries, cold-worked or stressed regions (a bent nail rusts at the bend), and above all **differential aeration** — a region with poorer oxygen access becomes anodic relative to one with better access. This is why rust starts under washers, inside crevices, and beneath a lifting paint edge, which are exactly the places that look most protected.

Differential aeration is counter-intuitive and worth thirty seconds: the sheltered spot corrodes *because* it is sheltered.

The reference table

Standard reduction potentials, acidic, versus SHE:

Half-reaction	E° (V)	Note for teaching
$\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}$	-2.37	Water heater rods
$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	-1.66	Thermodynamically wild, kinetically tame
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$	-0.76	Galvanizing, hull anodes
$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$	-0.44	The metal at risk
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	0.00	Reference
$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	+0.34	Noble relative to iron
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$	+0.80	Session 3
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	+1.23	The real cathodic reaction in wet corrosion

That last row is the one instructors most often skip and most need. In an open jar, the copper electrode is not reducing Cu^{2+} — there is barely any Cu^{2+} present. It is a surface on which **dissolved oxygen** is reduced. The Cu^{2+}/Cu line still predicts the right *direction* for the couple, which is why we use it, but it is not the reaction taking place, and it is a large part of why measured voltages disagree with predictions.

Part B coaching — the predict-then-measure gap

Collect the predictions before you hand out a single multimeter. The pedagogical value is entirely in the commitment.

Expected behaviour:

Pair	Predicted	Typically measured	Comment
Zn-Cu	+1.10 V	0.9–1.1 V	The well-behaved one
Fe-Cu	+0.78 V	0.4–0.7 V	Reasonable

Pair	Predicted	Typically measured	Comment
Zn-Fe	+0.32 V	0.2–0.4 V	Small but correctly signed
Mg-Cu	+2.71 V	1.4–1.7 V	Large shortfall; Mg also reacts directly with water
Al-Cu	+2.00 V	0.3–0.9 V, erratic	The teaching moment

Three reasons for the shortfall, in order of size:

1. **Wrong half-reaction at the cathode.** Oxygen reduction, not metal deposition. What students measure is a **mixed potential** — the compromise where the anodic and cathodic rates match — not a thermodynamic cell potential.
2. **Not standard conditions.** No 1 M metal ions anywhere; Nernst shifts everything.
3. **Surface films.** Passive oxides, corrosion product build-up, and area ratios.

Do not grade on closeness to E° . Grade on whether they got the **direction** right — which metal corrodes — and whether they can name a reason for the gap.

Aluminium is the lesson, not the outlier

Al is predicted second only to Mg and will usually read below iron. It carries a passive Al_2O_3 film a few nanometres thick that is adherent, insulating and self-repairing. This is why we build aircraft and window frames from a metal that is thermodynamically eager to burn.

Push the class to the general principle: **the table tells you what wants to happen, never how fast**. Everything students will meet later in kinetics, catalysis, and materials engineering lives in that gap. It is also the same distinction they met yesterday as overpotential — the same idea, wearing different clothes.

If someone sands the aluminium hard and immediately immerses it, they may briefly see a much larger voltage that then decays as the film reforms. That is worth stopping the room for.

The short-circuit current — do not skip it

Open-circuit voltage says whether corrosion is favoured. Current says how fast. This is the step that turns Day 5 from qualitative into a real measurement, and it reuses Faraday's law for the third time in three days.

$$\text{mass rate of Zn loss} = I \times 65.4 / (2F) \qquad 1 \text{ mA} \approx 11 \text{ g of zinc per year}$$

The reading decays over the first half-minute as the electrodes polarize. That decay is itself informative — corrosion current depends on how easily the cathodic reaction can be supplied with oxygen, which is why stirring or aerating speeds it up. Take a value at ~30 s and note that it is not stable.

Sizing a ship's anode is exactly this calculation performed on a hull. Say so.

Corrosion jars — narrative and failure modes

Jar	Setup	What they should see	Story
A	Fe alone	Moderate, even rust	Local cells on one metal
B	Fe + Cu in firm contact	Worse rust on the iron	Galvanic couple: Fe anode, Cu cathode

Jar	Setup	What they should see	Story
C	Fe + Zn in firm contact	Iron clean; zinc dull and powdery	Sacrificial protection

Failure mode, and it is the common one: the contact slips. No electron path, no couple, no difference between B and A. Wire the metals together tightly and check them on the day you set them up and again mid-week. A jar B that looks like jar A is not a subtle result, it is a disconnected circuit.

Second failure mode: jars filled to the brim. Fill them **half full** so the metals cross the waterline, where oxygen supply is best and the effect is most visible.

Have a photograph of a properly developed set of jars available as a backup. If this year's jars underperform, showing what they should look like and explaining why yours did not is a perfectly good lesson — and considerably better than pretending.

Chloride's role

Saltwater is the right choice here: it supplies ions, and chloride actively breaks down passive films, which is why marine environments are so aggressive and why road salt destroys cars. It is also exactly why chloride is banned from the derusting bath and from Session 4. Same property, opposite desirability. Making that explicit stops "no salt" sounding like an arbitrary safety rule.

Electrolytic derusting — the polarity point

Piece	Terminal	Role
Rusty object	– cathode	Reduced, cleaned
Sacrificial steel or graphite	+ anode	Oxidized, consumed, sludges
Electrolyte	Washing soda (or baking soda)	Alkaline, no chloride

Reversing the clips dissolves the workpiece. This is the cheapest, sharpest demonstration of the week's central idea and it costs nothing to state: same jar, same battery, same solution, opposite outcome, decided entirely by which wire went where.

Graphite anodes shed carbon and blacken the bath. Steel anodes produce more sludge but are cheaper and more robust. Stainless steel is sometimes suggested online and should be avoided — it can release chromium compounds into the bath.

Overnight practicalities

- **Not a 9 V alkaline.** ~500 mAh against a cell drawing a few hundred mA is roughly two hours. Use a mains supply or a 5 V USB brick. This is the most common reason a Session 5 reveal disappoints.
 - **Hydrogen all night.** Jars open, room ventilated, no ignition sources. A capstone question (§5 B2) has students calculate that an overnight cell can generate a couple of litres of H₂ — worth doing, because it converts the rule into arithmetic.
 - Secure the wiring so nothing can tip or short.
-

Misconceptions

Misconception	Correction
"Rust is a coating that lands on iron."	Rust is the iron, oxidized. Nothing arrived; electrons left.
"Copper attacks iron chemically on contact."	Copper does nothing. It provides a cathode surface, and the iron becomes the anode of a real cell.
"Zinc protects steel by sealing it."	It does seal, but it also sacrifices . That is why a scratch in galvanizing does not rust and a scratch in paint does.
"Any coating protects. Plating is plating."	A noble coating (silver, chrome) must be perfect — a breach makes a small anode under a large cathode and pits fast. A sacrificial coating protects through the breach.
"The measured voltage should match the table."	It should not. Non-standard conditions, oxygen reduction at the cathode, and surface films. The direction is what the table gets right.
"Aluminium is unreactive."	It is one of the most reactive metals in use. Its oxide film is what is unreactive.
"Derusting turns rust back into steel."	Some oxide is reduced; much is mechanically lifted by hydrogen. Lost metal is lost.
"Salt is the best electrolyte."	Best for corrosion demos, forbidden for anything driven — chlorine and pitting.
"Stainless steel can't corrode."	Its passive Cr_2O_3 film can be broken by chloride, and it pits in crevices.

Board plan

1. Treated nail vs control, held up. Say nothing for a beat.
2. $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$ and $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$. Write "rust **IS** the iron."
3. Activity series, left to right, large. Leave it up all session.
4. Jars A / B / C sketched, with the anode labelled in each.
5. Two-column table: rusting (spontaneous, Fe = anode) vs derusting (driven, Fe = cathode).
6. $1 \text{ mA} \approx 11 \text{ g Zn per year}$, circled.
7. The six experiments in a row, one arrow under all of them, for the closing.

Optional enrichment

- **Crevice corrosion and differential aeration** — why bolted joints and washers fail first.
- **Stainless steel passivation** — Cr_2O_3 , and why "stainless" fails in seawater and in crevices.
- **Impressed-current cathodic protection** — pipelines and jetties; the industrial version of the derusting cell.
- **Pourbaix diagrams for iron** — for a TA or a strong student who wants to know why pH matters.
- **Ferroxyl indicator** (potassium hexacyanoferrate(III) plus phenolphthalein), if you can get it for next year. It stains anodic sites blue and cathodic sites pink within **minutes**, on a bent nail or a Fe-Cu couple, and would let this session show corrosion happening live instead of relying on jars started days earlier. It is a standard school reagent; the rules are never to acidify it and never to heat it. This is the single best upgrade available to Session 5.

Session 5 — Preparation

Check this first, this morning

- [] **Are the derusting cells still powered?** Touch the supply, look for bubbling, confirm the nail is dark rather than orange. If the power died in the night, read the fallback below before class rather than during it.
 - [] **Is the control nail there, unpowered, in the same solution?** Without it there is no experiment.
 - [] **Can you find the before photos?** Have them open and ready to display.
-

Session 4 handoff (the critical dependency)

These cells must be running before students leave Session 4:

Item	Detail
Electrolyte	Washing soda preferred; baking soda acceptable — never NaCl
Cathode (-)	Rusty iron
Anode (+)	Sacrificial steel or graphite (not stainless)
Control	Equally rusty nail, same solution, no power
Power	Mains supply or 5 V USB brick. A 9 V alkaline is flat in about two hours
Ventilation	Jars open, room ventilated — these cells make hydrogen all night

If the reveal failed

It happens, usually because a battery died or a clip came loose. Options, in order of preference:

1. **Run a fast cell live.** Wire a fresh rusty nail at the highest safe current at the start of class and come back to it at the 60-minute mark. An hour at 0.5 A visibly changes a lightly rusted nail.
2. **Have a prepared example.** Keep one successfully derusted nail from your pilot run, sealed in a bag. Show it, and be straightforward that it is from the pilot.
3. **Teach the failure.** Ask the class to diagnose it: dead battery, loose clip, reversed polarity, too little electrolyte. Then check the anode — if it is heavily eaten, current definitely flowed, and the question becomes where the effect went. This is a genuinely good fifteen minutes and students respect it more than a smooth demo.

Whatever happens, write down what went wrong in `shared/teaching-log.md` while it is fresh.

Before class (instructor)

- [] Mix **5–10% saltwater** and label it "**CORROSION DEMOS ONLY — NOT FOR ELECTROLYSIS OR DERUSTING**"
- [] Physically separate the saltwater from anything left over from Session 4
- [] Sand all metal strips to bare metal, especially the **aluminium**, and keep them dry until use
- [] Confirm every multimeter has a **working mA range** — Part B3 needs it, and a blown fuse reads zero

- [] Pilot the metal pairs yourself and record what your saltwater actually gives; the numbers in instructor.md are typical, not guaranteed
- [] Check the corrosion jars: are the metal contacts still firm? A slipped wire makes jar B look like jar A
- [] Print E° tables with the **predictions column blank**
- [] Print capstone worksheets and week-comparison tables
- [] Locate photos of a ship's hull anode and a water-heater rod

Multi-day setup this session depends on

The corrosion jars need **four to five days** to develop. They belong at the end of Session 1, not here — see shared/multi-day-setup.md.

Jar	Setup
A	Iron nail alone in saltwater
B	Iron nail in firm contact with copper
C	Iron nail in firm contact with zinc, or a galvanized nail

Fill each jar **half full** so the metals cross the waterline. Keep them where students can look at them daily — the running commentary through the week is half the value.

If the jars were not started: set them up today anyway as a demonstration of method, and use the photograph fallback for the results. Then put a note in the teaching log so next year's Day 1 includes them.

Day-of timeline

When	Task
Before students	Verify cells still powered; disconnect nothing yet
0 min	Part A — reveal, with control and before-photo
15 min	Activity series, galvanic couples, sacrificial protection
30 min	Part B — predictions collected before meters are handed out
55 min	Part C — jars A / B / C
68 min	Part D — week comparison table
78 min	Capstone worksheet, then the closing
90 min	Collect worksheets; photograph everything for next year

Troubleshooting

Issue	Cause	Fix
Treated nail looks like the control	Power died, clip loose, or too little time	See the fallback options above
Nail is <i>worse</i> than when it started	Polarity reversed — it was the anode	Excellent teaching moment; show the class the wiring
Bath smells of chlorine	Salt was used	Stop, ventilate, dispose, remake with washing soda
Metal-pair voltage near zero	Metals touching in solution, or an oxide layer	Separate them; sand both
Aluminium reads lower than iron	Expected — passive oxide film	This is the lesson, not a fault
Voltages well below prediction	Non-standard conditions; oxygen reduction at the cathode	Expected. See instructor.md
Short-circuit current reads zero	Meter not on mA, or a blown fuse	Check the fuse
Jar B looks identical to jar A	Contact slipped — no electron path	Re-clamp; note it for next year
Jars barely rusted at all	Started too late, or filled too full	Use the fallback photo; start jars on Day 1 next year

Open questions / notes for next year

- Overnight power source used, and hours run:
- Was the reveal convincing?
- Measured voltages: Zn-Cu _ / **Fe-Cu** / Al-Cu _ / **Mg-Cu**
- Zn-Cu short-circuit current:
- Date the corrosion jars were started, and whether that was early enough:
- Which capstone questions worked, and which fell flat:
- Did the session finish on time, or did the reveal overrun?